

- 20**      **A**      Weight of oil drop = Electric force (vertically upwards) + Force due to air resistance

$$Mg = QE + kv,$$

$$\text{Rearranging, } v = \frac{Mg}{k} - \frac{QE}{k}$$

A graph of  $v$  against  $E$  will give a straight line of gradient,  $-\frac{Q}{k}$  and  $y$ -

**5**

intercept,  $\frac{Mg}{k}$